Problem

Problem · Constructing a regular pentagon with ruler and compass. Let $\omega = e^{2i\pi/5}$ and $u = \omega + \frac{1}{\omega}$.

1. Prove that $u = 2\cos\frac{2\pi}{5}$.
2. Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$, then, dividing by ω^2, that $u^2 + u - 1 = 0$.
3. Deduce that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.
4. In an orthonormal frame, draw the unit circle and place $I(-\frac{1}{2}, 0)$ and $J(0, 1)$.
   (a) Compute IJ.
   (b) The circle with centre I through J meets the positive half of the horizontal axis at K. Prove that the abscissa of K is $2\cos\frac{2\pi}{5}$.
   (c) Deduce a ruler-and-compass construction of the vertices of a regular pentagon inscribed in the unit circle.